

BHUVANIKA S

Aspiring Electronics Engineer

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EDUCATION

2023-2027

BE Electronics and Communication engineering

Kongunadu College of Engineering and Technolgy

CGPA: 8.31

SKILLS

CORE TECHNICAL SKILLS:

- VLSI
- Embedded system/ IoT
- FPGA
- RTL Design
- PCB

PROGRAMMING LANGUAGES:

- Java
- Python
- C
- C++

SKILLS:

- Problem Solving
- Leadership
- Event Organization
- Communication

TOOLS

- Modelsim
- Multisim
- VivadoXilinx
- VS code
- Matlab

LANGUAGES

- French
- English
- Tamil

PROFILE SUMMARY

Highly motivated B.E. Electronics and Communication Engineering graduate with a passion for learning and innovation. Eager to leverage academic knowledge and project experience to contribute to a challenging technical role while continuously growing professionally.

PROJECTS

Titile: SmartHealthMonitoring and Decision Support System

Technologies Used: Verilog HDL, Vivado.

Description:

Developed a Verilog-based health monitoring system to analyze multiple health parameters and detect abnormal conditions.
Implemented decision-making modules to provide health status evaluation and preventive recommendations.

Title: VLSI Implementation of ECG Prediction

Technologies Used: Verilog HDL, Vivado, FPGA.

Description:

Designed a hardware-efficient ECG signal denoising and R-peak detection system using Verilog HDL.
Optimized the design for low-area utilization and high-speed FPGA implementation.

Title: Fire and Gas Detection Bot

Technologies Used: IoT, ESP8266/ESP32, MQ-2 Sensor

Description:

Developed a mobile robot to detect fire and hazardous gas leaks using MQ-2 sensors.
Integrated Wi-Fi connectivity to send real-time alerts and improve emergency response.

Title: Autonomous Navigation Robot

Technologies Used: Arduino, Ultrasonic Sensor, Motor Driver

Description:

Designed an autonomous robot with obstacle avoidance using ultrasonic sensors.
Implemented navigation algorithms for real-time path detection and movement control.

INTERNSHIP

Embedded Software Intern

ImbedSoftware Company | June 2025 – July 2025

- Developed and tested Embedded C programs for microcontroller-based applications.
- Performed firmware debugging and validation to improve software reliability.
- Worked with hardware peripherals and sensor interfacing.
- Collaborated with engineers in troubleshooting embedded systems.
- Reduced debugging time through systematic firmware testing.